

Task 4

Setup and Use a Firewall on Windows

Objective

To configure and test basic firewall rules using Windows Defender Firewall in order to allow or block network traffic on a system.

Tools Used

- Windows Defender Firewall with Advanced Security
 - Windows PowerShell
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Deliverables

- Screenshot of Windows Defender Firewall showing the custom rule blocking TCP port 23.
 - Screenshot of PowerShell output showing the result of the connection test to port 23.
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1. Opening the Firewall Configuration Tool

1. Click on the **Start** menu.
2. Search for “**Windows Defender Firewall with Advanced Security**”.
3. Open the application.

This console allows us to view and configure inbound and outbound firewall rules for the local computer.

2. Listing Current Firewall Rules

1. In the left pane of the console, click **Inbound Rules**.
2. The main pane displays all existing inbound rules, including their name, profile, enable/disable status, action (Allow/Block), program, and protocol.

This step gives an overview of which applications and services are currently allowed or blocked by the firewall.

3. Creating a Rule to Block Inbound Traffic on Port 23 (Telnet)

1. With **Inbound Rules** selected, go to the right-hand **Actions** panel and click **New Rule...**

2. In the **Rule Type** screen, select **Port** and click **Next**.
3. In the **Protocol and Ports** screen:
 - Select **TCP**.
 - Select **Specific local ports** and enter: 23.
 - Click **Next**.
4. In the **Action** screen, choose **Block the connection** and click **Next**.
5. In the **Profile** screen, ensure all three are selected:
 - **Domain**
 - **Private**
 - **Public**Then click **Next**.
6. In the **Name** screen:
 - Name: Block Telnet Port 23 (Lab)
 - (Optional) Description: Rule to block inbound TCP traffic on port 23 for lab task.
 - Click **Finish**.

The new rule now appears in the Inbound Rules list and is enabled with the action set to **Block**.

4. Testing the Firewall Rule Using PowerShell

To verify that the firewall is controlling access to port 23, a connection test is performed from PowerShell.

1. Open **Windows PowerShell**.
2. Run the following command:

```
Test-NetConnection -ComputerName 127.0.0.1 -Port 23
```

- ComputerName 127.0.0.1 refers to the local machine (loopback address).
- -Port 23 specifies the Telnet port that we blocked.

The output shows details such as:

- ComputerName
- RemoteAddress
- RemotePort
- PingSucceeded
- TcpTestSucceeded

If `TcpTestSucceeded` is **False**, it indicates that a TCP connection to port 23 could not be established, which is consistent with the blocking rule.

If it is **True**, the test shows the port is reachable; however, the important point for the lab is that the command is used to verify whether firewall rules are affecting connectivity to that port.

A screenshot of this result is taken as evidence of testing.

5. Removing the Test Block Rule to Restore Original State

After testing is completed, the system is restored to its previous firewall configuration.

1. Return to **Windows Defender Firewall with Advanced Security**.
2. Click **Inbound Rules**.
3. Locate the rule named **"Block Telnet Port 23 (Lab)"**.
4. Right-click the rule and select **Delete**.
5. Confirm deletion when prompted.

This removes the custom block rule and returns the firewall to its original state.

6. Commands and GUI Steps Used

GUI Steps (Windows Defender Firewall):

- Opened *Windows Defender Firewall with Advanced Security*.
- Viewed existing **Inbound Rules**.
- Created a new inbound rule of type **Port**, protocol **TCP**, specific port **23**, action **Block the connection**, applied to **Domain, Private, and Public** profiles.
- Deleted the custom rule after testing.

PowerShell Command Used:

```
Test-NetConnection -ComputerName 127.0.0.1 -Port 23
```

This command is used to test connectivity to a specific TCP port on a target machine and verify the effect of the firewall rule.

7. Summary – How the Firewall Filters Traffic

A firewall is a security mechanism that monitors and controls network traffic based on predefined security rules. When an inbound packet arrives at the system, Windows Defender Firewall checks the packet against its rule set:

- If there is a rule that **blocks** traffic for that protocol and port (for example, TCP port 23), the firewall **drops the packet** and the connection is not established.

- If there is an **allow** rule or no blocking rule, the traffic may be permitted according to the default policy.

By creating a rule to block TCP port 23 and then testing connectivity, we see how the firewall can be used to restrict access to specific services. Removing the rule restores normal behavior. This demonstrates how host-based firewalls reduce the attack surface by only allowing required network traffic and blocking unnecessary or potentially dangerous ports.

8. Conclusion

In this task, Windows Defender Firewall was used to:

- Review existing inbound firewall rules.
- Create a custom rule to block inbound TCP traffic on port 23.
- Test connectivity using Test-NetConnection from PowerShell to observe the effect of the firewall rule.
- Remove the custom rule to restore the original configuration.